

Processor Design Project [CESE4040] (Q4 2025-2026)

Georgi Gaydadjiev and Motta Taouil

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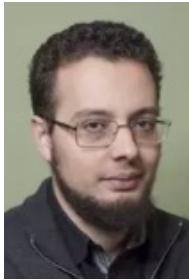
Carlo Galuzzi, Stephan Wong and Mahmood Naderan-Tahan

April 2026

Summary of the Basic Information

- Course Code: **CESE4040**
- **Lecturers:** Motta and Georgi
- **Weekly labs coordinated by:** Carlo, Stephan and Mahmood
- **Teaching Assistants:** (see next slide but also Brightspace)
- **Team Channels (per group):** (will be created as the groups are formed)
- **Examination:** large project performed in **groups of four** students
- **Lecturing method:** **one** introductory lecture and weekly lab sessions
- **Book:** N/A (see BrightSpace/Webpage for more information)
- **Preliminary knowledge:** Computer Engineering (CESE4130) and Modern Computer Architecture (CESE4085). Some of you may also have followed Computer Arithmetic (CESE4035) and Compiler Construction (CS4555) in Q2.
- **BRING YOUR OWN LAPTOP!**
- **Additional reading:** recommended literature per topic will be provided

The CESE4040 (2025) team



Motta



Georgi

Overall course strategy
and organization



Carlo



Mahmood



Stephan

Weekly lab sessions
organization and
project monitoring

TA team 2025:



Ignacio



Ismail



Fouwad



Pantazis



Manos



Konstantinos

Course structure CESE 4040 (2026)

- **One lecture on Wednesday, April 22**
 - Covering organization / main topics of the project
 - Additional information needed for your progress
 - Slides, manuals, etc. on BrightSpace
- **Weekly lab sessions**
 - Every Friday afternoon from 13:45 to 17:45 in EWI-Tellegen Hall rooms 1, 2 and 3
 - TAs and Labs coordinators will guide you
- **One short intermediate report** (end of week 2, May 8)
 - You will receive feedback one week later
- **Your RISCY_AES+LLVM design files along with your final presentation**
 - Delivered at the end of week 8 (June 12, 2026)
- **Group presentation / demo and Q&A**
 - Schedule a slot of **60min** in weeks 25 and 26 with your TA

Grading Scheme (2026)

This is **your project** and we all are here to assist you but **not** for doing the project for you. Your final grade will depend on these parts:

- The **minimal requirements** (working AES encryption instructions and the LLVM loop-unrolling pass)
- The **technical merit** (reflecting the innovation of your extensions and their implementation quality)
- **Final presentation, demo and oral group exam**
- **Your individual contribution** (rated by your project partners, and us, e.g., at the **Q&A**) [-1 ... +1]

Grading Scheme (2026) (cont)

The grading will be based on the **final examination** using the **following criteria**: *(the precise weights of those will be communicated later)*

1. **Suitability of the proposed solution** to given problem;
2. **Efficiency of implementation** of chosen solution;
3. **Technical Depth** of results obtained **and** your **analysis**;
4. **Final presentation** (clarity, group performance).

The above determines **the group grade**.

Subsequently, **personalised differentiation** among students within the group will be applied. We will use: the **Buddy Check method** (similar to CESE4030 in Q3) and your **individual performance** as observed **during the final examination** (Q&A). This can lead to a **maximal +/- 1** (grade) point difference from the group grade above.

Processor Design Project organization

- **The Goal:** *setup, plan and execute a large project to extend a RISC-V-based computer system and its associated LLVM toolchain with hardware and software support for an industry standard encryption algorithm (AES). You will also propose additional improvements, implement, validate and evaluate those.*
- Seven (7) half-day labs (on Friday afternoon) for guidance and feedback
 - Content and additional information on BrightSpace/Webpage
- You will work **in groups of four (4)**
- Please enroll on BrightSpace by the **end of the TODAY** (Wed, April 22) as after that you will be assigned automatically, and the groups will be compressed/compacted to contain the required number of students (4).
- **Support:**
 - joining the lab sessions is really recommended to pass this course
 - asking questions is highly encouraged
 - you can always send a **Teams message** as described earlier
 - please note, that after sending your question some time might be needed to provide a good answer

Processor Design Project Phases

- We build upon the labs in Q1 (CESE4130) and Q3 (CESE4085)
- You have been working on one extremely simple and another very complex RISC-V processors (Q1)
- In Q3 you also extended LLVM with the AES custom instruction for the middle rounds
- The above forms the basics for this course
- In PDP you will extend and improve a more realistic core named RISCY with AES hardware and work on real LLVM optimisations
- There will be **two** major PDP **phases** this year

Processor Design Project Phases (2)

- **First Phase** (weeks one and two)
 - Setup your project (RISCY + AES in software) and your design environments
 - Analyze the bottlenecks of the unmodified base system (AES software implementation on RISCY)
 - Decide on your intended improvements and your evaluation strategy/methodology
 - Study the state of the art (papers related to your intended improvements, you do not have to implement all of them)
 - Plan carefully your activities for the Second Phase
 - Summarize the above in your **intermediate report (due end of week two, Friday, May 8)**



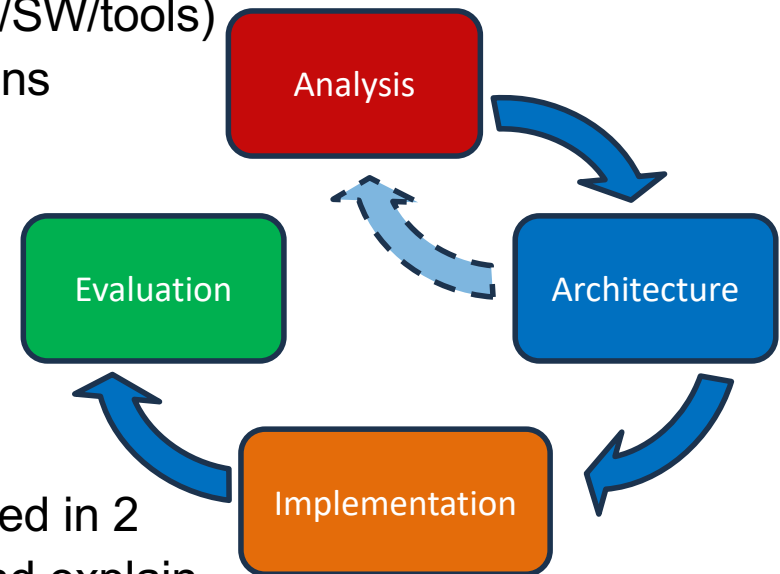
Processor Design Project Phases (3)

Second Phase (weeks three to eight)

- Implement and debug your aes32esi and aes32esmi extensions for RISCY
- Implement and debug an LLVM loop unrolling pass for the aes32esmi instruction
- Evaluate the advantages of the above hardware and software extensions and summarize your findings
 - the comparison is between the base RISCY and AES in software system and your RISCY+AES in hardware (RISCY_AES) and loop unrolling in LLVM
- Implement your additional improvements (as defined in your intermediate report) on the top of the above RISCY_AES and/or LLVM
- Evaluate the improvements and compare against the **state-of-the art**
 - **Note:** the baseline to compare against was defined and agreed upon in your intermediate report
- Summarize all of the above in your **final presentation (due end of week eight)**
- Deliver your sources (RTL and LLVM) and your final presentation on time
- Present your work, provide a demo and answer our questions
- **PASS THE COURSE!**

What are we trying to do here?

- **Step 1: Analyse “Code”**
 - Study/profile code, measure time taken
 - Measure memory requirements and working set sizes
 - Understand the workload requirements and the system limitations
- **Step 2: “Architect” your Solution**
 - Think of options for improvement, e.g., partitioning, u-arch ext., compiler opt.
 - Consider the entire system and workload (HW/SW/tools)
 - Study the state of the art of the identified options
 - Form a hypothesis for targeted “speedup”*
 - Explore “speedup” options (e.g., model/argue)
 - [#1 Best “Architecture”+“Speedup” baseline]
- **Step 3: Implement your Best Candidate**
 - Implement and debug your improvements
- **Step 4: Evaluate your implementation**
 - Measure and check against the baseline defined in 2
 - Analyse your findings (match/under/over...) and explain
- **Result: “Speedup” validated (or studied at least)**



* “speedup” is the improvement you will implement and validate

TA Support Guidelines



What to expect from TAs

Valid support requests

- 1 I have an idea implementing X to support my idea. What do you think?
 - **Valid!** – Seeking feedback on approach
- 2 Server connection is slow or I am not able to connect to the server.
 - **Valid** – Technical issue related to infrastructure



TAs will help with conceptual questions and technical issues



NOT to expect from TAs

Invalid support requests

- 1 When I run simulation/ synthesis/ compilation/ execution, I get an error message XYZ.
 - **Invalid** – part of the learning goal is to come up with ideas on how to debug and solve errors.
- 2 I am trying to setup LLVM/Vivado locally, but I get an error message.
 - **Invalid** – The server provides infrastructure to support your designs. Local installation, as well as any problems arising, is a personal adventure.



TAs won't debug your design or fix local setup issues

Corner cases

- In case you didn't follow/pass CESE4130 and CESE 4085, please contact Motta and Georgi asap
- Partial grades from the last year **can't be** used
 - **Repeating students** should preferably perform the entire course as the rest of the students
 - Please contact us in case of questions about this
- **It is important** to let your colleagues know way ahead of time if you will not be able to attend one of the lab sessions

The 2026 Schedule

Phase 1

Date	Time	Course code	Location(s)
week 17			
Wed 22 Apr	13:45 - 15:45	CESE4040	ME-Lecture Hall E - Robert Hooke (34.A-0-710)
Fri 24 Apr	13:45 - 17:45	CESE4040	EWI-Tellegen Hall practicumzaal 1 (36.Tellegen Hall practicumzaal 1) EWI-Tellegen Hall practicumzaal 2 (36.Tellegen Hall practicumzaal 2) EWI-Tellegen Hall practicumzaal 3 (36.Tellegen Hall practicumzaal 3)
week 18			
Fri 1 May	13:45 - 17:45	CESE4040	EWI-Tellegen Hall practicumzaal 1 (36.Tellegen Hall practicumzaal 1) EWI-Tellegen Hall practicumzaal 2 (36.Tellegen Hall practicumzaal 2) EWI-Tellegen Hall practicumzaal 3 (36.Tellegen Hall practicumzaal 3)
week 19			
Fri 8 May	13:45 - 17:45	CESE4040	EWI-Tellegen Hall practicumzaal 1 (36.Tellegen Hall practicumzaal 1) EWI-Tellegen Hall practicumzaal 2 (36.Tellegen Hall practicumzaal 2) EWI-Tellegen Hall practicumzaal 3 (36.Tellegen Hall practicumzaal 3)

Intermediate Report

Phase 2

Feedback on your Intermediate Report
(our target is one week later)

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Phase 2 (cont)

Date	Time	Course code	Location(s)
week 21			
Fri 22 May	13:45 - 17:45	CESE4040	EWI-Tellegen Hall practicumzaal 1 (36.Tellegen Hall practicumzaal 1) EWI-Tellegen Hall practicumzaal 2 (36.Tellegen Hall practicumzaal 2) EWI-Tellegen Hall practicumzaal 3 (36.Tellegen Hall practicumzaal 3)
week 22			
Fri 29 May	13:45 - 17:45	CESE4040	EWI-Tellegen Hall practicumzaal 1 (36.Tellegen Hall practicumzaal 1) EWI-Tellegen Hall practicumzaal 2 (36.Tellegen Hall practicumzaal 2) EWI-Tellegen Hall practicumzaal 3 (36.Tellegen Hall practicumzaal 3)
week 23			
Fri 5 Jun	13:45 - 17:45	CESE4040	EWI-Tellegen Hall practicumzaal 1 (36.Tellegen Hall practicumzaal 1) EWI-Tellegen Hall practicumzaal 2 (36.Tellegen Hall practicumzaal 2) EWI-Tellegen Hall practicumzaal 3 (36.Tellegen Hall practicumzaal 3)
week 24			
Fri 12 Jun	13:45 - 17:45	CESE4040	EWI-Tellegen Hall practicumzaal 1 (36.Tellegen Hall practicumzaal 1)

Working RISCY_AES and LLVM+AES compiler
and **your own** improvements and extensions
Final Presentation with evaluation of results

Weeks 25/26: Group presentations & demos of working design and Q&A (please schedule these with your lead-TA)

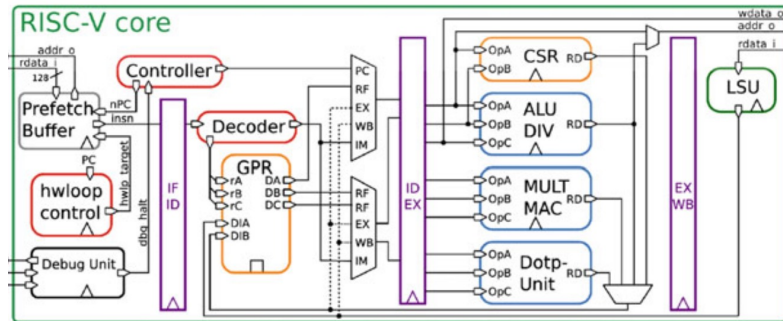
Some Additional Information

- ✓ We will use **only** Bright Space
- ✓ The course website: <https://cese.ewi.tudelft.nl/processor-design-project/index.html> is not up to date
 - ✓ You can use it as an additional reference only
- ✓ Additional information will be made available as we progress
- ✓ If you are really interested in learning, please ask questions!
- ✓ Plagiarism will not be tolerated in any form and shape!

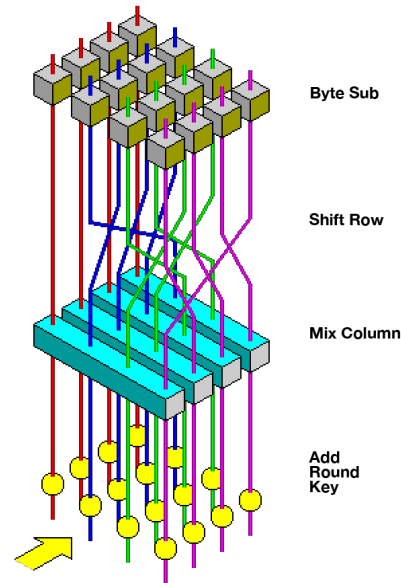
- ✓ Our plans are very ambitious in respect of the intended work, so good planning and progress monitoring is a key
- ✓ But you will learn a lot about computer design, RISC-V, AES hardware extensions and LLVM compiler optimizations

Questions about the PDP 2026 organisation

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